# Practical Training of Deep Networks

Normalisation, stabilisation, and diagnosis

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**Training is a coupled dynamical system**

**Initialisation recap: Xavier, He, and PyTorch defaults**

**Normalisation layers control activation scale**

**Batch normalisation**

**BatchNorm during training and evaluation**

**Batch size and data dependence**

**Layer normalisation**

**Initialisation and normalisation complement each other**

**Gradient clipping limits unstable updates**

**Learning-rate schedules and warm-up**

**Weight decay connects optimisation and regularisation**

**Training curves give diagnostic evidence**

**Inspecting activations and gradients**

**Dead and saturated units**

**A systematic failure-diagnosis workflow**